Part 2: Insights from Exploratory Data Analytics & Identification of Variable Relationships

1. Insights Gained from Exploratory Data Analytics (EDA)

Exploratory analysis of the Irish Property Price Register (2019–2024) revealed several important market characteristics:

- Property prices show a **strong right-skewed distribution**, with the median price significantly lower than the average price.
- There are **substantial regional disparities**, with Dublin and surrounding counties exhibiting much higher prices and sales volumes than rural counties.
- **Second-hand properties dominate the market**, accounting for approximately **82% of all transactions**, indicating constrained new housing supply.
- Clear **seasonal and quarterly patterns** are visible, with:
    - Sales peaking in **December**
    - Lowest activity in **January**
    - Prices generally increasing from **Q1 to Q4**
- Counties with higher sales activity tend to experience **lower affordability and higher price volatility**.

These findings naturally lead to deeper analytical questions about **drivers, relationships, and future risks**, which form the basis of Part 2.

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2. Analytical Questions Derived from EDA

Based on the exploratory findings, users of this dataset (policy makers and property investors) are likely to ask the following questions:

1. Which factors most strongly influence **property prices** across Ireland?
2. Does **sales volume** contribute to **price growth and affordability pressure**?
3. Which counties are at the **highest risk of affordability stress**?
4. How does **property condition** influence **price volatility and market risk**?

To answer these questions, four key sets of variable relationships were identified.

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3. Identified Variable Relationship Sets

3.1 Relationship Set 1: Property Price Determinants

Variables Used (from Tableau):

- Real Price
- Median Price
- County Clean
- Property Condition

- Year

Relationship Explored:

This relationship examines how **location, property condition, and time** influence property prices.

EDA Evidence:

- Dublin (€582,327) has an average price nearly four times higher than counties such as Leitrim (€151,944).
- New properties consistently show higher median and average prices than second-hand properties.

Why This Relationship Matters:

Understanding price drivers enables:

- **Policy makers** to design location-specific housing interventions
- **Investors** to estimate expected price levels in different counties

This relationship supports **price prediction and forecasting analysis**.

3.2 Relationship Set 2: Sales Volume vs Price Growth

Variables Used:

- Total Sales
- YoY Median Price Growth %
- Year
- Year Month

Relationship Explored:

This relationship analyses whether **higher transaction activity** leads to **increased price growth**.

EDA Evidence:

- Dublin accounts for over **102,000 sales**, far exceeding other counties.
- Years with higher transaction volumes coincide with **higher YoY median price growth**, particularly around 2021–2022.

Why This Relationship Matters:

- High sales pressure often indicates **demand-driven price inflation**
- Useful for identifying **overheating markets**

This relationship supports **demand–price interaction analysis**.

3.3 Relationship Set 3: Affordability vs Market Pressure (County Level)

Variables Used:

- Affordability Index

- Affordability Index (County)
- Total Sales
- County Clean

Relationship Explored:

This relationship investigates how **sales intensity impacts housing affordability** at county level.

EDA Evidence:

- Counties with high sales volumes (e.g., Dublin, Kildare, Wicklow) show **lower affordability index values**
- Rural counties have higher affordability but lower transaction activity

Why This Relationship Matters:

- Identifies counties under **affordability stress**
- Supports **targeted housing policy and planning decisions**

This relationship enables **risk scoring and affordability classification**

3.4 Relationship Set 4: Property Condition vs Price Volatility

Variables Used:

- Real Price
- Property Condition
- County Clean
- Year

Relationship Explored:

This relationship analyses whether **new or second-hand properties** experience greater **price variability** across counties.

EDA Evidence:

- Box plot analysis shows **wider price dispersion in urban counties**
- Second-hand properties display greater variability due to mixed property age and quality

Why This Relationship Matters:

- High volatility increases **investment risk**
- Helps investors assess **market stability**
- Helps policy makers identify **unstable housing markets**

This relationship supports **market risk and volatility analysis**.

3.5 Summary of Identified Relationships

Relationship	Key Variables	Insight Type
Price Determinants	Price, County, Property Condition, Year	Price Prediction
Sales vs Price Growth	Sales Volume, YoY Growth, Time	Market Pressure
Affordability Risk	Affordability Index, Sales Volume	Policy Risk
Price Volatility	Price, Property Condition, County	Investment Risk

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4. Predictive Dashboards Overview:

4.1 Predictive Dashboard 1 – County Median Price Forecast

Target Audience

- Government housing policy analysts
- Real estate investors and property developers

Purpose

This dashboard presents a **county-level forecast of median residential property prices** in Ireland, based on historical transaction data from 2019 to 2024. It uses **time-series forecasting** to estimate future price trends and highlight regional differences in expected market behaviour.

Information Presented

- Historical and **forecasted median property prices** by county
- **Confidence intervals** to reflect prediction uncertainty
- Interactive filters for **County** and **Property Condition (New vs Second-Hand)**

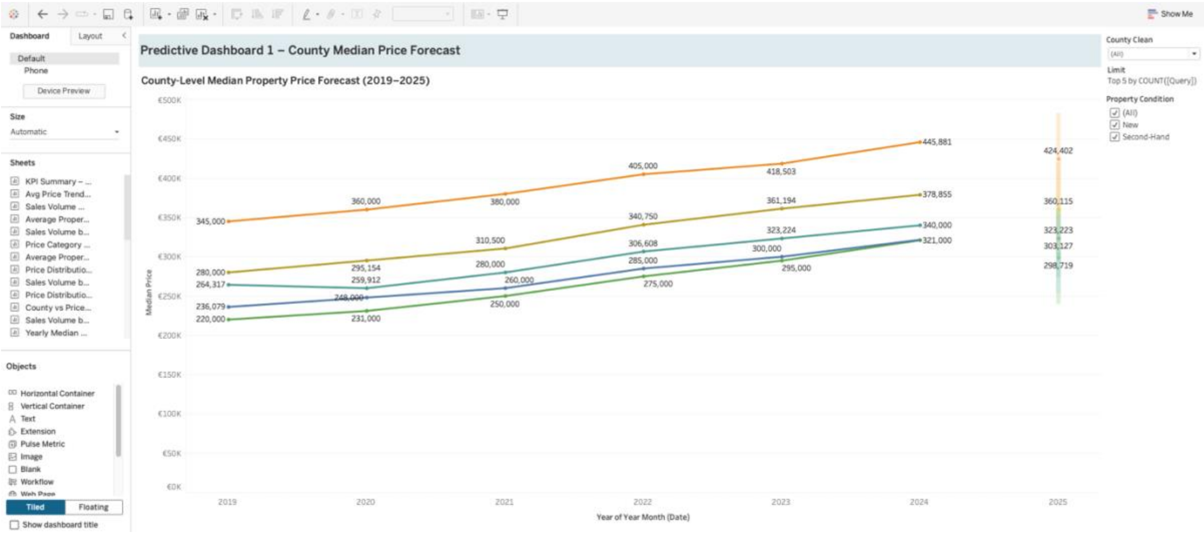
Key Insights

- Median property prices are forecasted to **continue rising across all major counties**
- Dublin maintains the **highest projected price levels**, followed by Wicklow and Kildare
- Growth trends are consistent, indicating **sustained demand rather than short-term volatility**

Actionable Insights & Decision-Making Value

- **Policy makers** can anticipate future affordability pressures and plan housing interventions accordingly
- **Investors and developers** can identify counties with strong price growth potential for long-term investment
- Forecast confidence bands support **risk-aware planning and scenario analysis**

This dashboard enhances **evidence-based decision-making** by translating historical market behaviour into forward-looking price expectations.



4.2 Predictive Dashboard 2 – Sales Volume vs Price Growth

Target Audience

- Housing policymakers, market analysts, real estate investors.

Purpose & Information Presented

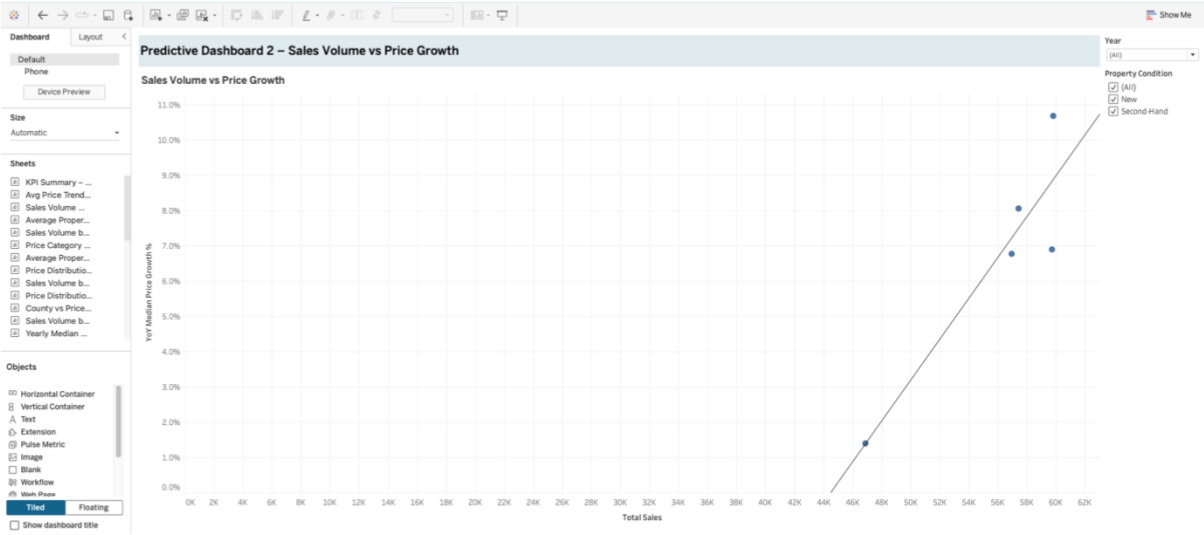
This dashboard examines the relationship between housing transaction volumes and year-on-year median price growth in Ireland. It visualises how changes in sales activity influence price growth using a regression-based predictive model.

Key Insights

The analysis shows a positive relationship between total sales volume and price growth. Years with higher transaction activity (e.g., 2021–2022) are associated with significantly higher YoY median price growth, indicating demand-driven price pressure.

Actionable Insights & Decisions

This dashboard helps identify periods of market overheating. Policymakers can use it to trigger affordability or supply-side interventions, while investors and planners can anticipate price acceleration during high-demand phases, supporting informed timing and risk-management decisions.



4.3 Predictive Dashboard 3 – Affordability vs Market Pressure

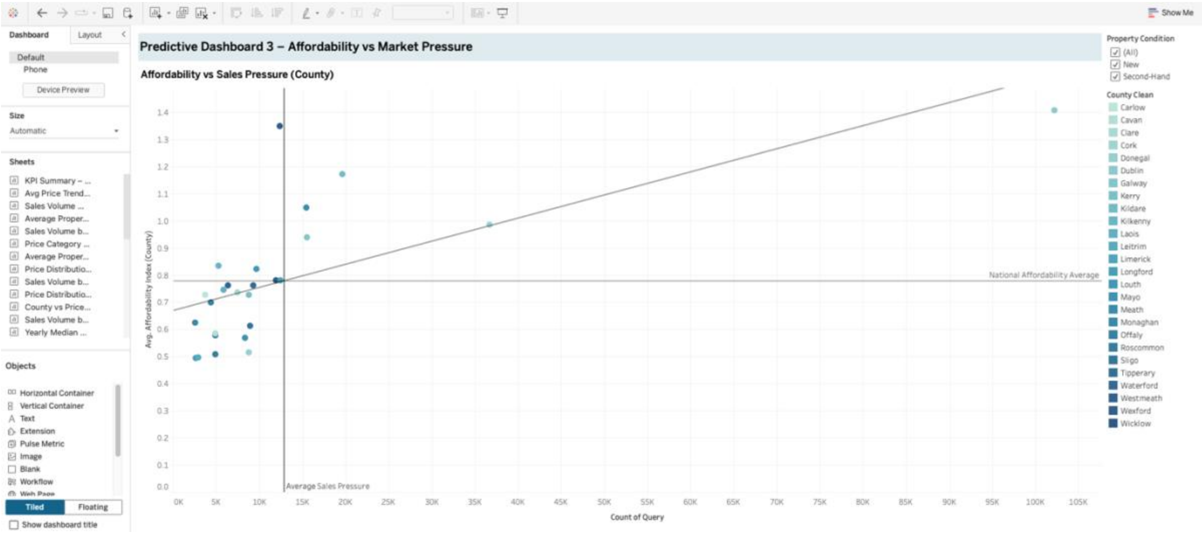
Dashboard Description (marks-ready)

Target Audience
Housing policymakers, local authorities, and urban planners.

Purpose & Information Presented
This dashboard analyses the relationship between county-level housing affordability and market pressure, measured through transaction volumes. A linear regression model is used to quantify how increased sales activity relates to affordability levels across counties.

Actionable Insights
Counties with high sales pressure and below-average affordability are identified as high-risk areas. These locations may require targeted housing interventions such as affordability schemes, supply expansion, or planning controls.

Decision Support
The dashboard supports evidence-based housing policy by enabling decision-makers to prioritise counties under affordability stress and allocate resources more effectively.



4.4 Predictive Dashboard 4 – Price Volatility Risk by Property Condition

Target Audience

- Property investors, housing analysts, policy makers, and financial institutions assessing market risk and stability across Irish counties.

Purpose

This dashboard evaluates housing market risk by analysing **price volatility (IQR)** across counties, segmented by **property condition (new vs second-hand)**. It identifies regions and property types with unstable pricing behaviour.

Key Insights

- Urban counties exhibit **higher price volatility**, indicating greater market risk.
- **Second-hand properties** consistently show higher volatility than new builds, reflecting heterogeneity in age, condition, and location.
- Rural counties demonstrate **lower volatility**, suggesting more stable and predictable pricing patterns.

Decision Support

- Enables investors to assess **risk-adjusted investment strategies**.
- Supports policy makers in identifying **unstable or overheated housing markets**.
- Assists planners in targeting **volatility mitigation and housing supply interventions**.


Show Me
Dashboard
Layout
Default
Phone
Device Preview
Size
Automatic
Sheets
KPI Summary – ...
Avg Price Trend...
Sales Volume ...
Average Proper...
Sales Volume b...
Price Category ...
Average Proper...
Price Distributio...
Sales Volume b...
Price Distributio...
County vs Price...
Sales Volume b...
Yearly Median ...
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Blank
Workflow
Tiled
Floating
Show dashboard title
Predictive Dashboard 4 – Price Volatility Risk
Price Volatility Index by Property Condition
County Clean
IQR Volatility
€250K
€200K
€150K
€100K
€50K
€0K
Dublin
Wicklow
Galway
Cork
Kilkenny
Limerick
Kildare
Kerry
Sligo
Waterford
Clare
Monaghan
Westmeath
Mayo
Tipperary
Wexford
Louth
Roscommon
Offaly
Meath
Cavan
Donegal
Carlow
Laois
Leitrim
Longford
- Higher IQR values indicate greater price volatility and market risk.
- Second-hand properties typically show higher volatility than new builds, particularly in urban counties.
County Clean
(All)
Property Condition
(All)
New
Second-Hand
Year
(All)
Property Condition
New
Second-Hand